

Jayasri Suresh Vani

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EXECUTIVE SUMMARY

Software Engineer with 2+ years of experience designing, developing, and deploying scalable full-stack applications and AI/ML systems across healthcare, fintech, and education domains. Proficient in Python, Java, and JavaScript with expertise in building robust backend services, RESTful APIs, and cloud-native architectures. Demonstrated ability to optimize database performance, implement CI/CD pipelines, and integrate machine learning models into production environments. Strong foundation in data structures, algorithms, and software design patterns, with a proven track record of delivering high-quality, maintainable code that meets business requirements.

EDUCATION

Purdue University

Master of Science in Computer Science (GPA: 3.9/4.0)

Anna University - Easwari Engineering College

Bachelor of Technology in Information Technology (GPA: 4.0/4.0)

Fort Wayne, Indiana

Aug 2024 - May 2026

Tamil Nadu, India

Sep 2020 - May 2024

SKILLS

- **Programming Languages:** Python, Java, TypeScript, JavaScript
- **AI/ML & Data Science:** Deep Learning, NLP, Predictive Modeling, Computer Vision, Transformers, Sequence Models, Scikit-learn, TensorFlow, PyTorch, RAG, LangChain
- **Frameworks:** React, Next.js, Angular, NestJS, Flask, Django
- **Databases & Cloud:** MySQL, PostgreSQL, MongoDB, Firebase, Supabase, Docker, Prisma
- **Tools:** Git, GitHub, REST APIs, MediaPipe

WORK EXPERIENCE

AI/ML Research & Development Intern | RAVEN AI | July 2025 - Present | New York, NY

- Developed NLP-driven behavioral scoring system for medical training evaluation across 5 communication dimensions.
- Benchmarked LLM evaluation pipelines (GPT-4o, Qwen3, Llama3) using prompt engineering and automated metrics (MAE, MSE, R²).
- Generated 250+ synthetic doctor-patient conversations using contextual filtering and patient-name injection for dataset expansion.

Technical Support Engineer | PURDUE UNIVERSITY | Dec 2024 - Present | Fort Wayne, IN

- Resolved 200+ IT support tickets involving network connectivity, hardware issues, and system access across university departments.
- Maintained and monitored 100+ university systems, ensuring high availability and reduced downtime.
- Collaborated with cross-functional IT teams to troubleshoot and resolve escalated issues efficiently.

Data Analyst & ML Intern | TECHVOLT PVT. LTD | Feb 2023 - Nov 2023 | Coimbatore, India

- Conducted EDA using Pandas, NumPy, and Seaborn to identify trends, anomalies, and actionable insights.
- Built diagnostic ML models (Logistic Regression, Naive Bayes, Decision Trees) achieving 85%+ accuracy using Scikit-learn and cross-validation.
- Applied transfer learning and data augmentation to fine-tune pre-trained models, improving predictive performance and robustness.

Software Developer Intern | OMEGA BINARIES | Apr 2023 - May 2023 | Chennai, India

- Architected and implemented a Java Swing desktop application using MVC design pattern with separate controller, model, and view layers, achieving 40% latency reduction and 25% faster UI responsiveness through optimized event handling and multithreading.
- Designed MySQL database schema with normalized tables and implemented AES-256 encryption for credential storage, supporting concurrent authentication for 100+ users with ACID transaction guarantees.
- Integrated MFS100 biometric SDK via JNI wrapper, implementing custom exception handling and callback mechanisms to enable real-time fingerprint verification and role-based access control (RBAC).

PROJECTS

CampusCrew - Centralized Service Platform

Jayasri2021/CampusCrew

- Created a service exchange platform connecting 500+ Purdue students, featuring a React front-end with role-based access and a Flask back-end with real-time Supabase integration. Optimized query latency by 40% and ensured reliability with Cypress/Pytest testing.

Auction-Ease: Online Auction Platform

Jayasri2021/Auction-Ease-Project

- Built and deployed a scalable auction platform supporting 1,000+ bids/min with real-time React interfaces and Java Spring Boot APIs, leveraging ETL pipelines, optimized concurrency control, and load-balanced REST services to reduce response time by 35%.

Adaptive Learning Management System

Jayasri2021/lms_project

- Structured a role-based LMS with secure authentication, automated quiz evaluation, and OLAP-driven analytics for 500+ students; Improved scalability using Docker containers and Prisma ORM for efficient database management.

MoodTunes -<https://moodtunes-423d2.web.app>

Jayasri2021/Music-Playlist-Gneneration-model

- Built a full-stack ML app classifying user emotions with 87% accuracy to generate personalized playlists; deployed on GCP Cloud Run with secure authentication and API integration, and hosted a Logistic Regression model on HuggingFace Space using Gradio SDK.

CERTIFICATIONS

- **Google Cloud Badge:** *Introduction to Generative AI*
- **Kaggle:** *Into to Machine Learning, Intermediate Machine Learning*
- **Udemy:** *Flutter & Dart, Django4 & Python Fullstack Developer*